

**NE585**  
**NUCLEAR FUEL CYCLES**  
**Front end of the nuclear fuel cycle**  
**3**

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# Learning objectives

Demonstrating how the nuclear fuel cycle is an holistic system

Explaining why the nuclear fuel cycle is what it is

Apply principle of SWU

Model a nominal solvent extraction system

Model enrichment processes

Demonstrate principles of enrichment and conversion

Some of Chapter 4 in the book

Check out the [Pandora's promise movie](#) if it's still on Netflix

There's a lot of pictures here

# Learning nodes

## **Nuclear fuel cycle overview**

**Locations of current domestic fleet**

**Plant closures**

**Nuclear power around the world**

**Planned construction**

**Components of the nuclear fuel cycle**

Mining

Milling

Conversion

Enrichment

Separative Work Unit (SWU)

Fuel fabrication

## **Macro reactor operation**

CP-1

Hanford B reactor

EBR-I

EBR-II

TREAT

Military use

## **Reactor concepts**

Generations

PWR

BWR

CANDU

## **Aqueous reprocessing**

# More learning nodes

## **Enrichment**

History

Processing

Tails

## **Conversion**

Processing

## **Nuclear fuel cycle overview**

# Nuclear fuel cycle

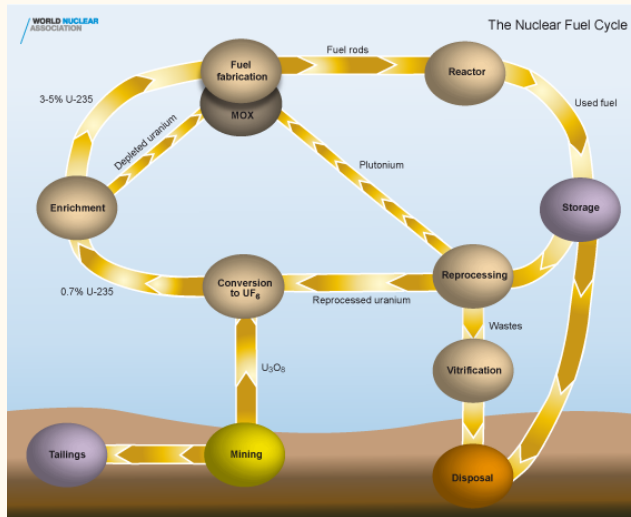
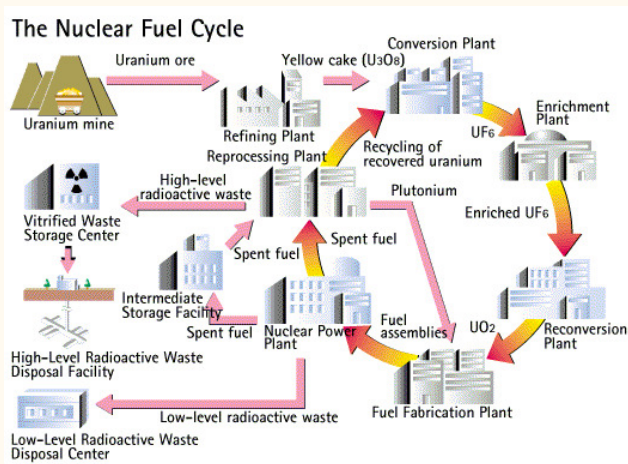


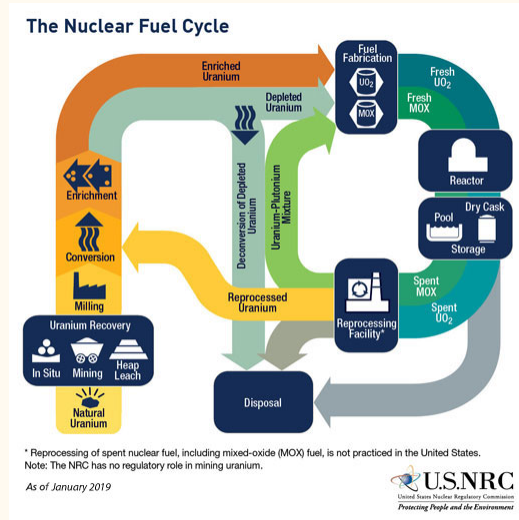
Figure 1. Nuclear fuel cycle

# Nuclear fuel cycle



**Figure 2.** Nuclear fuel cycle

# Nuclear fuel cycle



**Figure 3.** Nuclear fuel cycle



# Look at the fuel cycle as systems engineering

Mining

Making the fuel (saw Areva plant in Richland)

What a reactor does and the different kinds

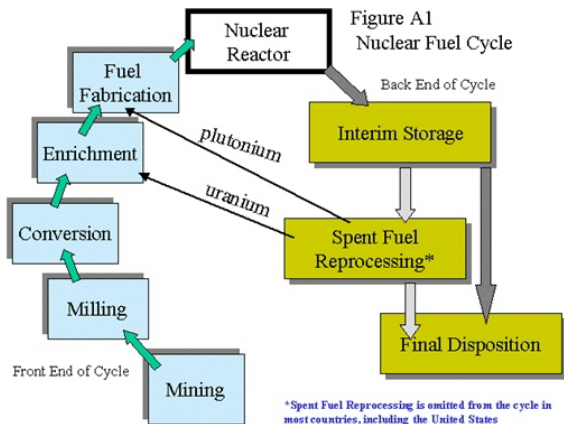
Recycling (what kinds and how)

Back-end management (storage and disposal)

The fuel cycle has a lot of technical but institutional issues too

How are they functionally dependent?

# The fuel used in the commercial reactors is $UO_2$ or $(Pu + U)O_2$



**Front end** – Making the fuel and putting it in the reactor

**Back end** – What happens when you take it out of the reactor

**Figure 4.** Nuclear fuel cycle

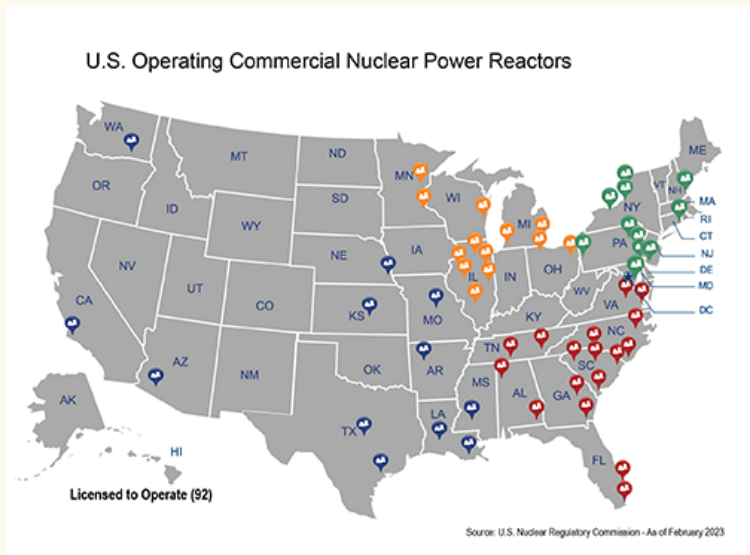
## **Locations of current domestic fleet**

## Nuclear Power Plant sites



### Figure 5. Nuclear Power Plant sites

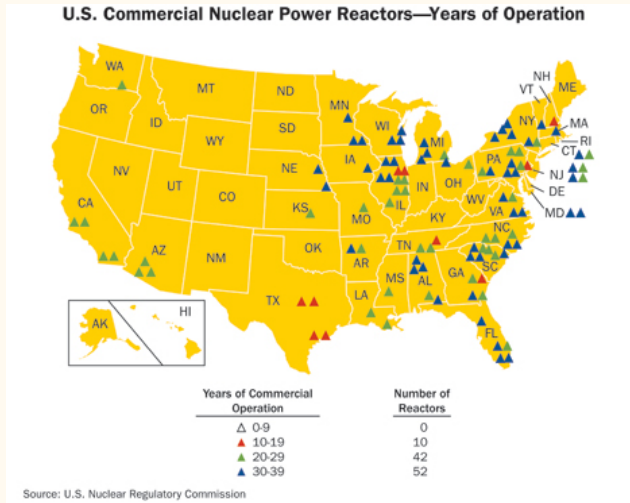
# Currently operating NPPs



**Figure 6.** Currently operating NPPs

**NRC list of power reactor units**

# NPP years of operation



**Figure 7.** NPP years of operation

**Plant closures list on the OER**



# Recent plant closures

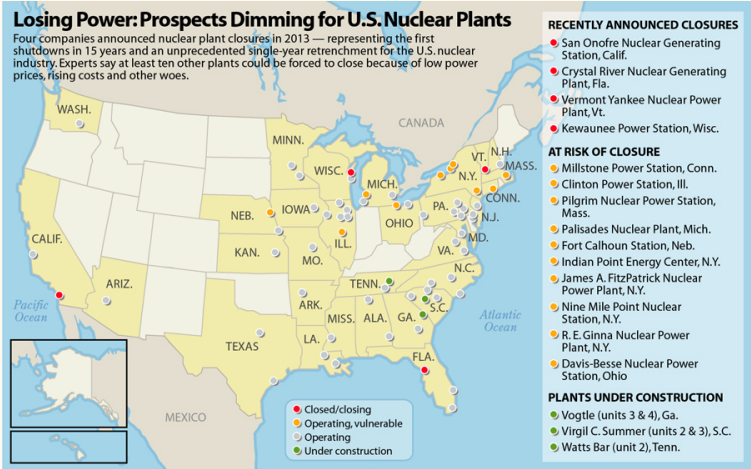


Figure 8. Recent plant closures

**Let's look at other nations**

# NPPs in Europe

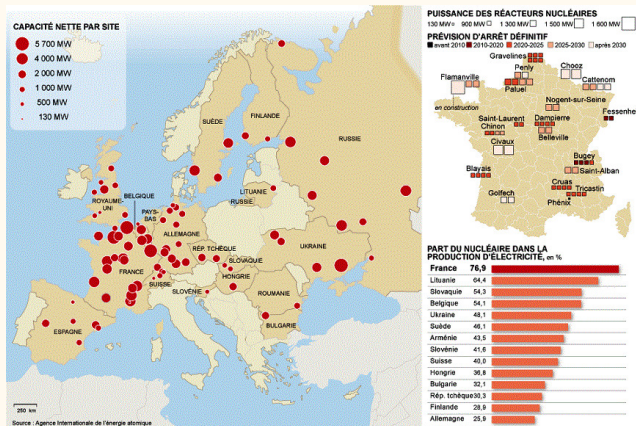


Figure 9. NPPs in Europe

# NPPs in Japan

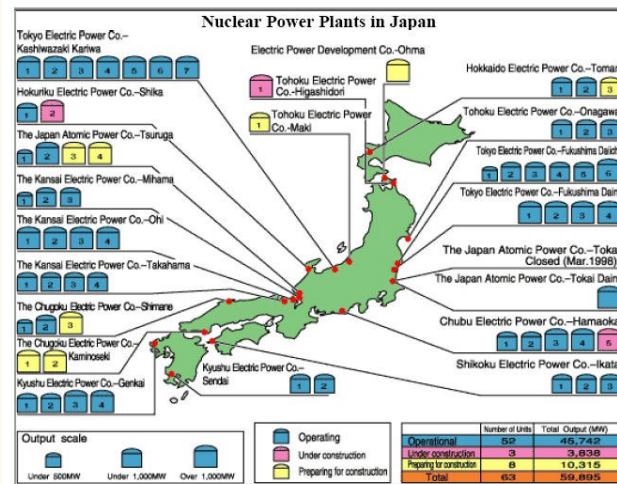


Figure 10. NPPs in Japan

# Where are they going to dispose of everything?



**Figure 11.** NPPs in Japan

# Same problem with Korea

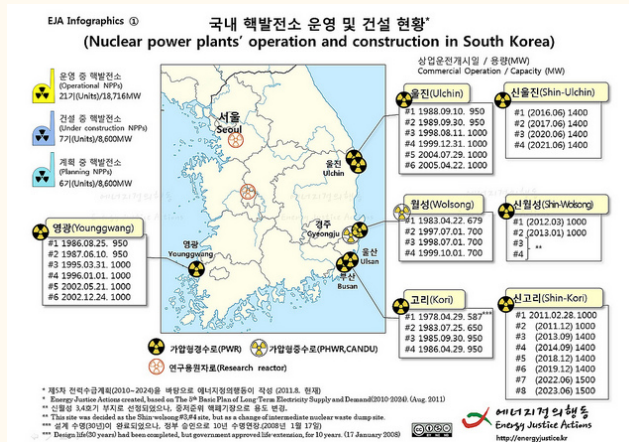


Figure 12. NPPs in Korea

# NPPs in Asia



Figure 13. NPPs in Asia

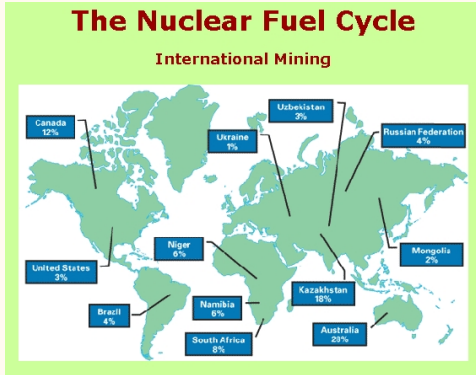
## **Plans For New Reactors Worldwide from the World Nuclear Association**



## **Components of the nuclear fuel cycle**

**Mining**

# We have to find uranium first



## $U_3O_8$ Nuclear Fuel Price Indicators

Extract the ore from the ground

What do you do if you don't have it?

Check out Australia

**Milling**

# Uranium ore is refined by milling

The ore contains  $UO_2$ ,  $UO_3$  and other heavy metal oxides

Usually using solvents to leach it from the ground

Then it is chemically treated to obtain 'yellowcake'

Lots of contaminated scrap is produced though (mill tailings)

Usually dumped as sludge in ponds (lots)

Radioisotopes (radon), heavy metals, chemicals, etc.

We have plenty of uranium resources

But what if we decide it is too harmful to mine?

**Conversion**

## $U_3O_8$ is converted to $UF_6$ prior to enrichment

Mined U contains  $^{234}U$ ,  $^{235}U$ ,  $^{238}U$ , .0054%, .72%

Because it is a gas at not so high temperature which is easier for enrichment

**Enrichment**



**$UF_6$  gas is then enriched with  $^{235}U$**

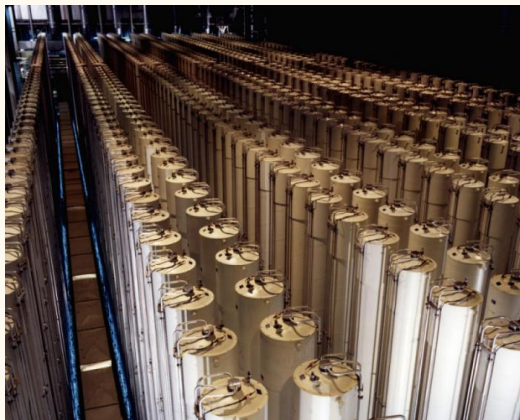
Enriched to 3% to 5%

The byproduct is depleted uranium DU

Used for armor and bullets

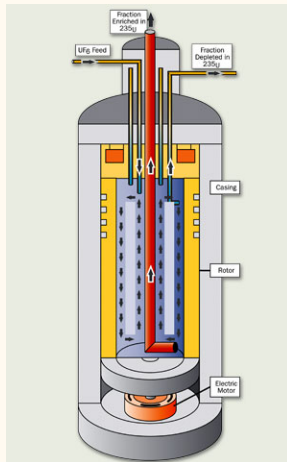
Any idea why?

# Enrichment uses centrifuge cascades



**Figure 14.** Centrifuge cascades

# Enrichment takes advantage of thermodynamics



**Figure 15.** Centrifuge process

## **Separative Work Unit (SWU)**

# SWU is the to enrich uranium from natural abundance to reactor/weapon grade

Dumb acronyms are why people hate nuclear

OER [documents](#) on SWU

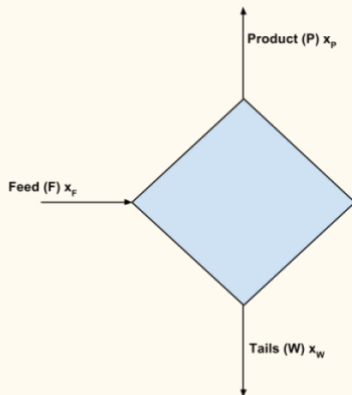
$$x_F = 0.0072 \quad (1)$$

$$x_P = 0.03 - 0.05 \quad (2)$$

$$x_W = 0.0002 - 0.0003 \quad (3)$$

Amount of feed for target product increases linearly with  $x_P$  by mass conservation

# Enrichment is just based on mass conservation



$$F = P + W \quad (4)$$

$$F \cdot x_F = P \cdot x_P + W \cdot x_W \quad (5)$$

**Figure 16.** Mass conservation

# SWU measures thermodynamic work to separate isotopes

SWU has units of kg because of course it does

Separation potential is a measure of differences in Gibbs free energy

$$SWU \equiv [P \cdot V(x_P) + W \cdot V(x_W)] - F \cdot V(x_F) \quad (6)$$

$$V(x_i) \equiv (2x_i - 1) \cdot \ln \frac{x_i}{1 - x_i} \quad (7)$$

$$\frac{SWU}{P} = V(x_P) + \frac{W}{P} V(x_W) - \frac{F}{P} V(x_F) \quad (8)$$

The derivation is based on entropy changes

Change in entropy in [binary mixture of gases \(paper in OER\)](#)

# What stands out?

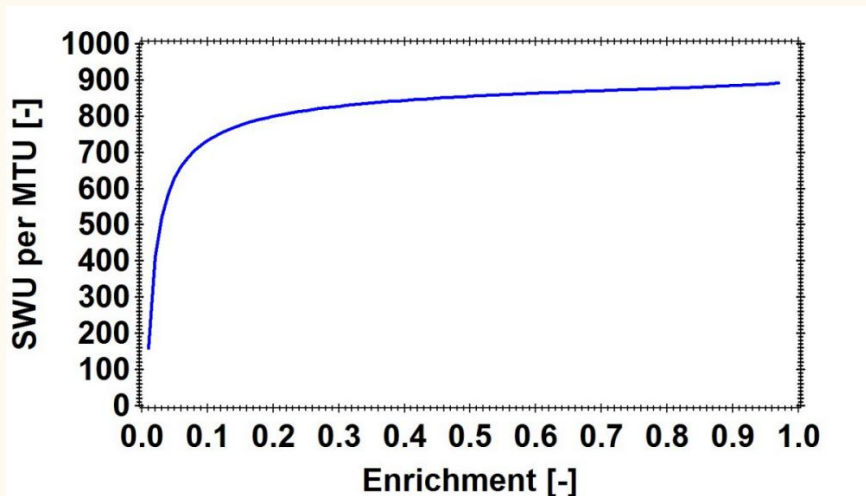


Figure 17. SWU curve varies with enrichment



## **Fuel Fabrication**

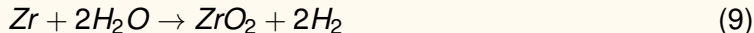
# The fuel fabrication process (LWRs) makes ceramic $UO_2$ pellets

Why ceramic?

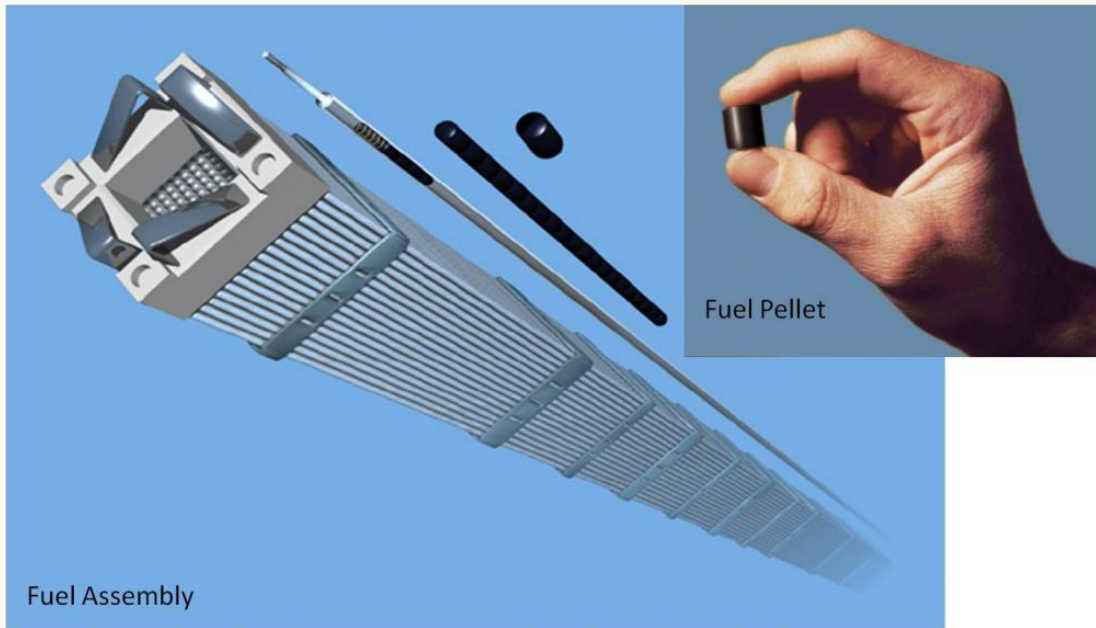
Is there anything better?

Fuel rods contain a Zr-based alloy

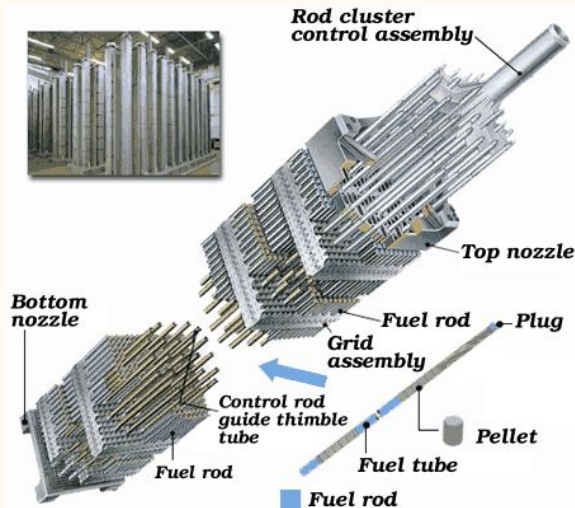
Paper on generating hydrogen gas



That's kind of what happened at [Fukushima \(article on Nature\)](#)



# How much energy do you get from one atom of uranium versus carbon?



**Figure 18.** Uranium fuel pellet and assembly

# Assemblies are about 12 feet tall

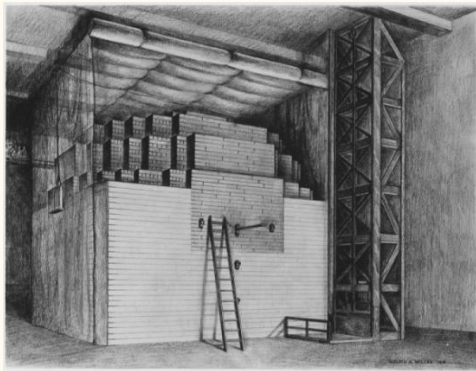


Figure 19. BWR assembly

## **Macro reactor operation**

## **Chicago Pile-1 (CP-1)**

# The first man made reactor was constructed at the University of Chicago in 1942



**Figure 20.** Chicago Pile-1

University of Chicago videos on the OER

Made of uranium and graphite blocks

Cadmium coated rods

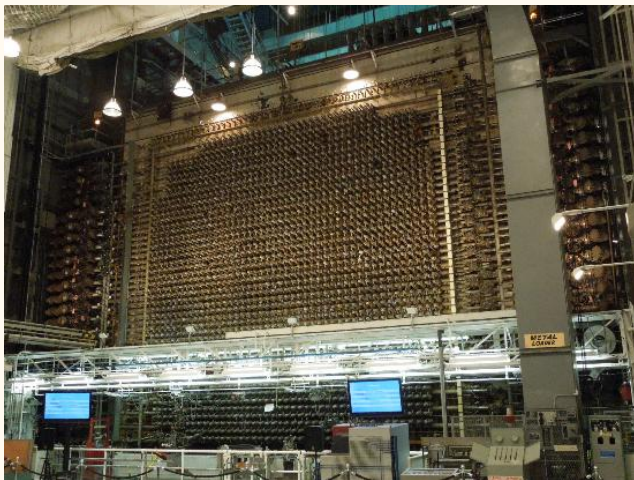
And I have a piece of the graphite



**Let's look at pictures of the Hanford B reactor**

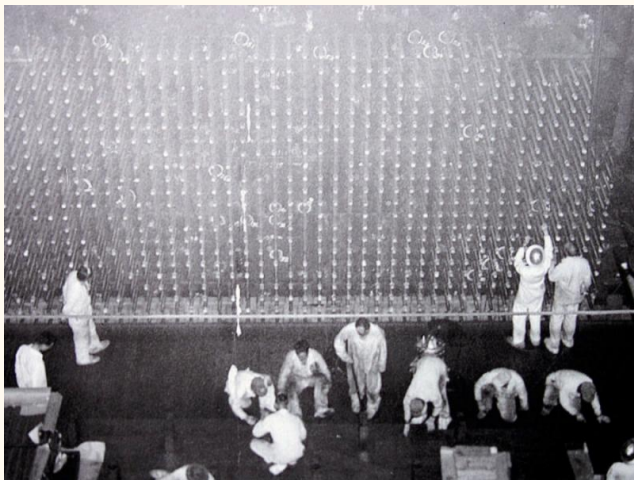
**Same design as CP1**

# Hanford B Reactor



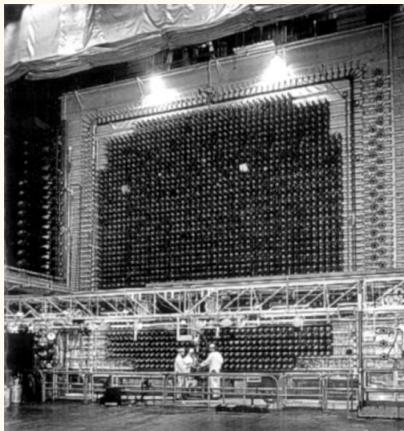
**Figure 21.** Hanford B Reactor

# Hanford B Reactor



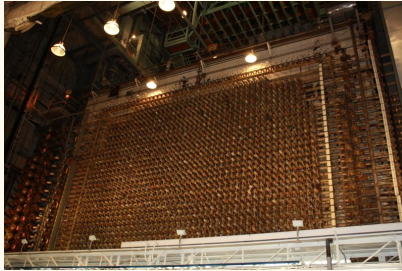
**Figure 22.** Hanford B Reactor

# Hanford B Reactor



**Figure 23.** Hanford B Reactor

# Hanford B Reactor



**Figure 24.** Hanford B Reactor

# Hanford B Reactor



**Figure 25.** Hanford B Reactor

# Hanford B Reactor



Figure 26. Hanford B Reactor



# Hanford B Reactor



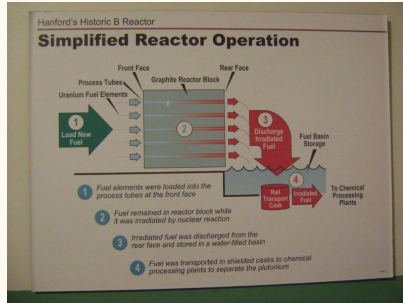
**Figure 27.** Hanford B Reactor

# Hanford B Reactor



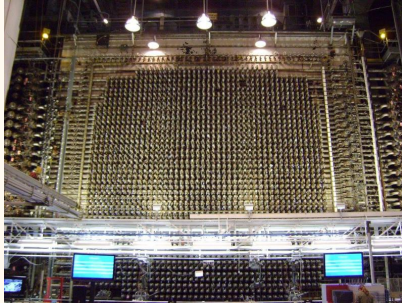
**Figure 28.** Hanford B Reactor

# Hanford B Reactor



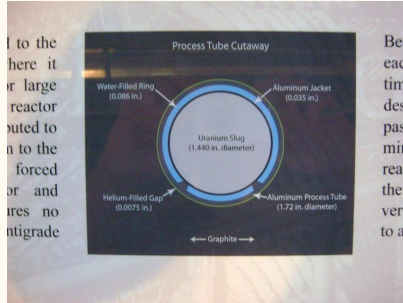
**Figure 29.** Hanford B Reactor

# Hanford B Reactor



**Figure 30.** Hanford B Reactor

# Hanford B Reactor



**Figure 31.** Hanford B Reactor

# Hanford B Reactor



**Figure 32.** Hanford B Reactor

# Hanford B Reactor



**Figure 33.** Hanford B Reactor

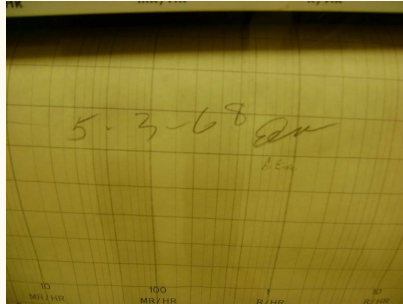
# Hanford B Reactor



**Figure 34.** Hanford B Reactor



# Hanford B Reactor



**Figure 35.** Hanford B Reactor

**Natural reactor**

# A natural reactor actually happened about 2 billion years ago

Fuel cycle on the OER

Neutronics on the OER

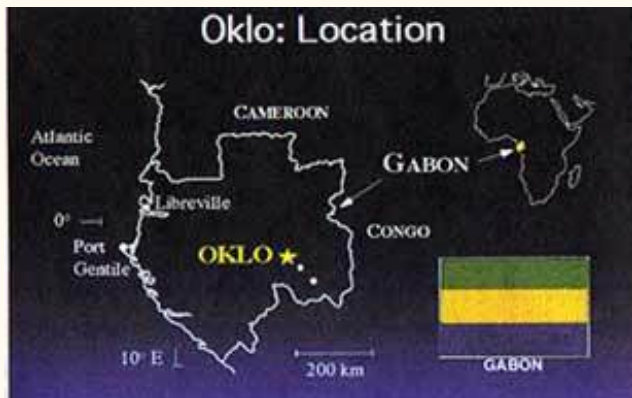
16 sites operated for about  $10^5$  years at 100 kW thermal

French company found samples of 0.60% of  $^{235}\text{U}$

How much is there normally?

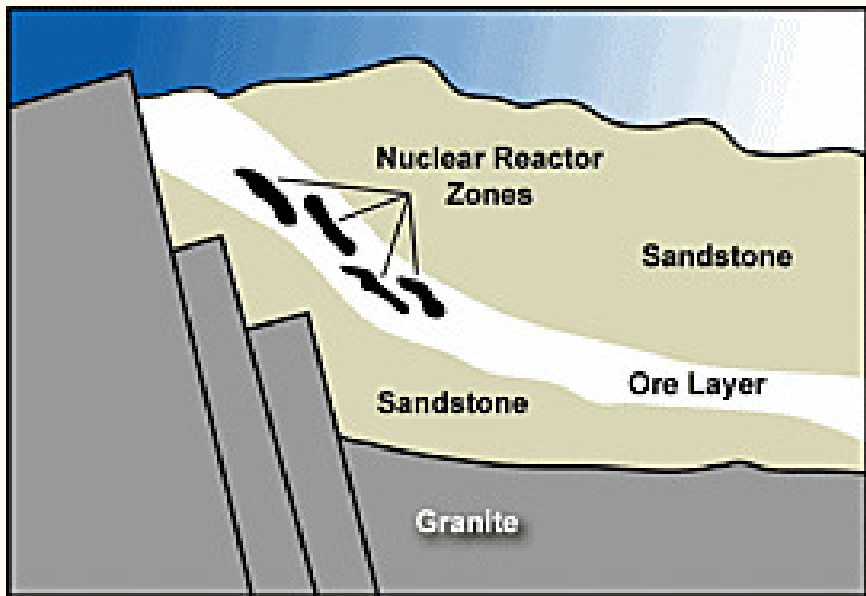
Uranium rich mineral deposits were infiltrated by groundwater which served as moderator

# Oklo location



**Figure 36.** Oklo location

# How the Oklo reactors worked



**Figure 37.** How the Oklo reactors worked

**EBR-I**

# EBR-I was the first reactor to generate electricity

In 1951 – now a museum, open to the public for free

Location of EBR-I

About 200 kW of electricity

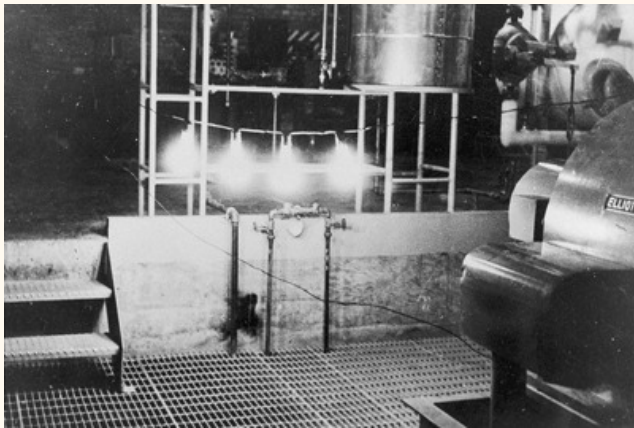
Also was a breeder and first to confirm Fermi's theory

'Fast' reactor, metal fuel and molten salt designs

Then the commercial reactors switched to ceramic and water

Why?

## EBR-I lit up 4 light bulbs



**Figure 38.** EBR-I lit up 4 light bulbs



**More pictures**

# EBR-I museum



**Figure 39.** EBR-I museum

# EBR-I museum



**Figure 40.** EBR-I museum

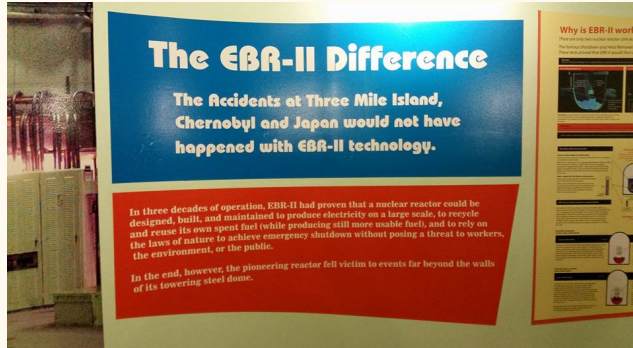


Figure 41. EBR-I museum

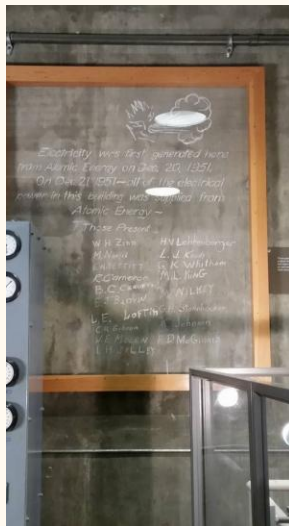


**Figure 42.** EBR-I museum



**Figure 43.** EBR-I museum

# EBR-I museum



### Figure 44. EBR-I museum



**Figure 45.** EBR-I museum



# EBR-I museum



**Figure 46.** EBR-I museum

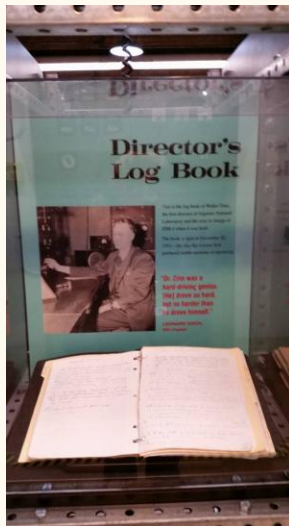
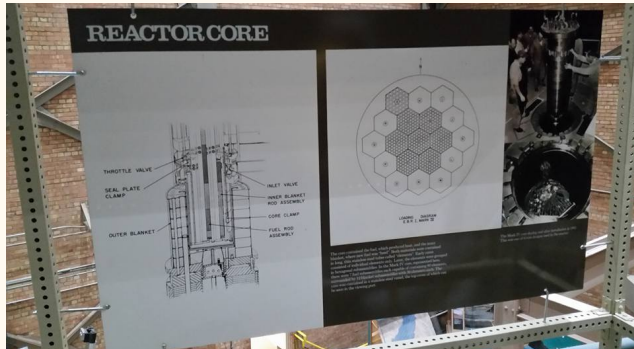


Figure 47. EBR-I museum

# EBR-I museum



**Figure 48.** EBR-I museum



**Figure 49.** EBR-I museum

# EBR-I museum



**Figure 50.** EBR-I museum

# EBR-I museum

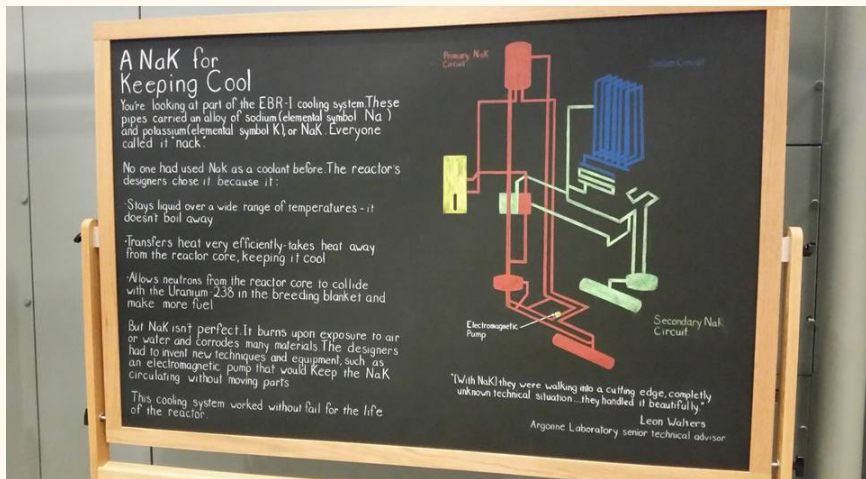


Figure 51. EBR-I museum



**Figure 52.** EBR-I museum

**EBR-II**



# EBR-II is an Sodium Fast Reactor (SFR) design for 62.5 MWth that operated from 1964 to 1994

19 MWe electricity operated for 30 years with no accidents

Engineering scale facility

Breeder reactor with onsite reprocessing (pyroprocessing)

[Famous test of passive safety systems video](#)

# EBR-II schematics

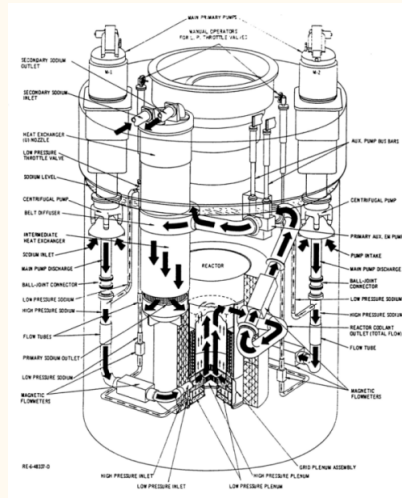


Figure 53. EBR-II schematics

**TREAT**

# TREAT is an air cooled, graphite moderated thermal reactor

Operation from 1959 to 1994

Transient reactor tests to simulate all sorts of reactor accidents

George Imel

Restarted and working on experiments!

Because of Fukushima, TREAT accident tolerant fuels

Everyone is excited

100 kW; transients up to 19 GW

# TREAT in person



**Figure 54.** TREAT

**Military use**

# Nuclear reactors are used on submarines and aircraft carriers

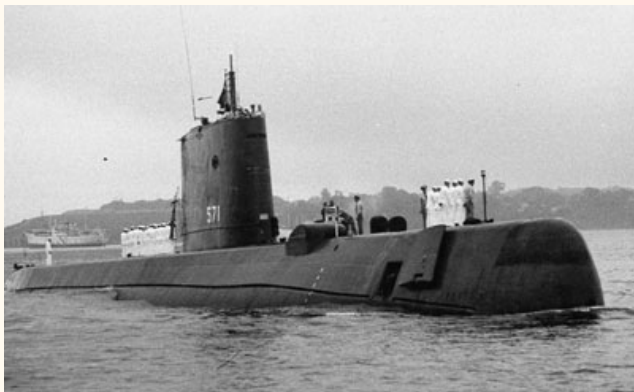
Newsreels on the Nautilus launch in 1954

25 year life span

Nimitz-class aircraft carriers have 2 reactors, about 100 MWe, 23 year life span

How about now?

# Nautilus submarine



**Figure 55.** Nautilus submarine



# Aircraft carriers have reactors too

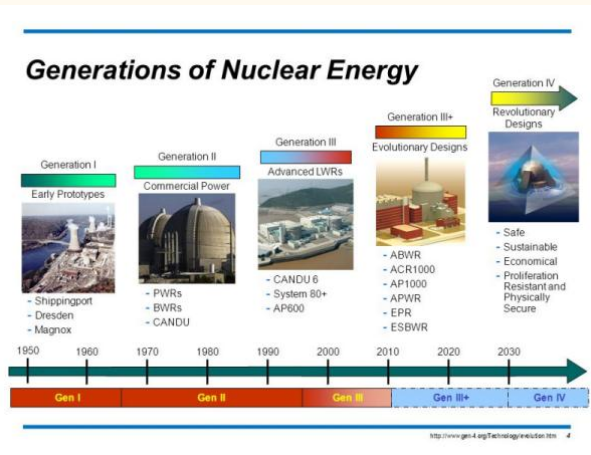


**Figure 56.** Aircraft carrier enters the danger zone. [Top Gun theme classic.](#)

## **Reactor concepts**

## **Generations**

# Reactors were first built in the 1950s



Gen IV is essentially based on Gen I

Different material for fuels and coolant

Moderators not needed for some

Passive safety systems

Proliferation issues – so they say

**Figure 57.** Reactor generations

**PWR**

# The PWR heats water but does not boil

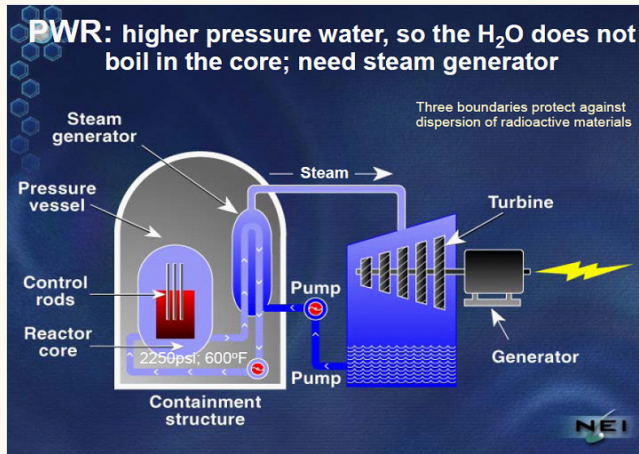


Figure 58. PWR

**BWR**

# The BWR does boil water in the reactor vessel

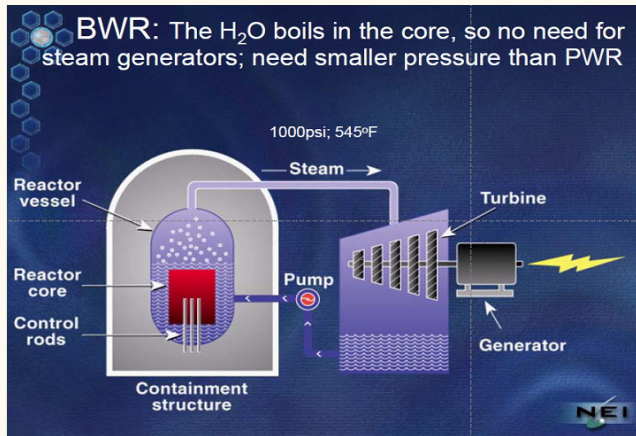


Figure 59. BWR



**CANDU**

# The CANDU design uses heavy water (moderator) and natural uranium (also thermal)

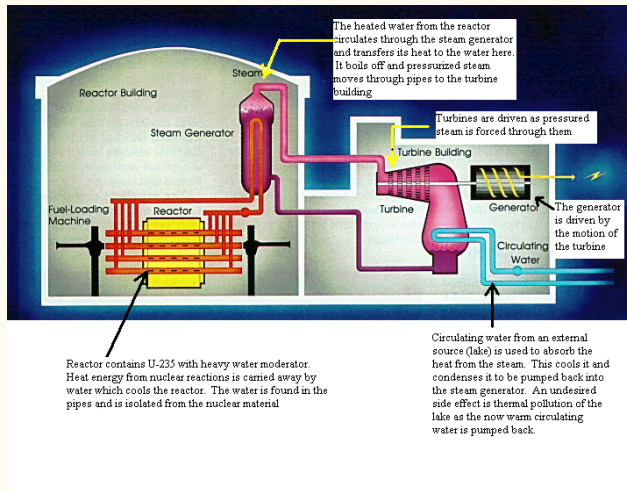


Figure 60. CANDU

## More on CANDU

Used in Canada and marketed to other nations

South Korea, China, India have some

PWR design (two loops)

The heavy water is less absorbing so natural U can be used

But also more collisions would be needed than in LWRs

The core would be bigger

But can use used fuel from LWRs and a Th cycle

So is there an advantage? Technical? Institutional?

**The CANDU design uses heavy water (moderator) and natural uranium (also thermal)**



**Figure 61.** Homer Simpson is bad at his job

## **Aqueous reprocessing**

# Aqueous recycling is the commercially available process for LWRs

[More information in the OER](#)

Terry Todd leading expert

Major nations are France, Japan, Russia, India

'Recycle' is better for the brand

About 1%  $^{235}\text{U}$  and 1% Pu in used fuel

The most common is PUREX

This uses an organic phosphate and nitric acid

Solvent extraction process where U and Pu are separated

Then created into MOX fuel

So for PUREX what is a big risk?

# PUREX flowsheet

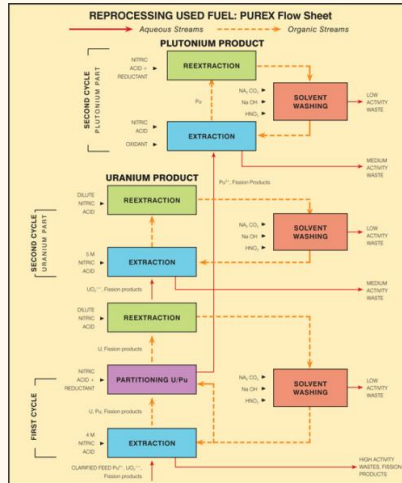


Figure 62. PUREX flowsheet

# U and Pu are chemically separated through organic solvent equilibrium extraction

U and Pu exist in a number of valence states

Because of different redox potentials, reduction or oxidation can be done to one without disturbing the other

Each exhibits different solubilities in organic solvent TBP

- (1) Chop up everything, remove FP gases, dissolve in aqueous solution
- (2) Separate U/Pu from fission products by dissolving in solvent
- (3) Reduce Pu to 3+ because it is highly insoluble in solvent (U 6+)
- (4) Back extract Pu into aqueous solution
- (5) Strip U from solvent into aqueous solution



# U and Pu are chemically separated through organic solvent equilibrium extraction

Aqueous solution is nitric acid

Does generate a lot of liquid waste

Mass transfer to solvent phase – extraction

Mass transfer to aqueous phase – “stripping”

# Mass transfer operates in a continuous, multistage, countercurrent mode

High separation factor

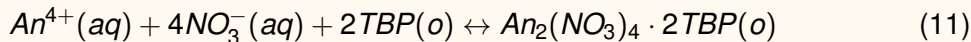
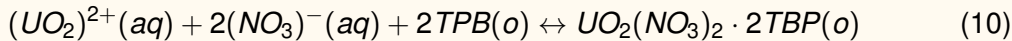
High processing rates

Streams flow countercurrently, so concentration is relatively constant

Solvent must be recycled because it's not cheap

First applied to extract Pu for weapons production in 1952 at Hanford site

## Metals in 4+ and 6+ are extracted



Pu (as 4+) co-extracted with U

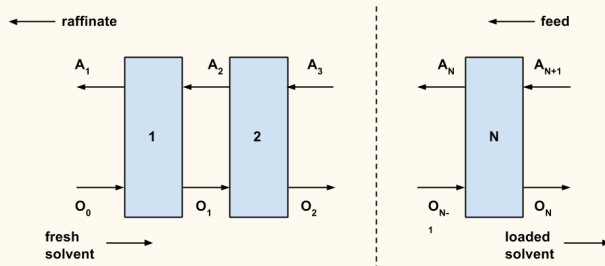
Anything 3+ or lower is not

Waste is vitrified into a borosilicate glass waste form for disposal

Well characterized material with 50+ years of scientific study

Very stable

# PUREX mass conservation



**Figure 63.** PUREX stage extraction

# Typical mass conservation is used to model solvent extraction

This is from Benedict

Distribution coefficient = organic/aqueous phase concentration at equilibrium

$$D \equiv \frac{y}{x} \quad (12)$$

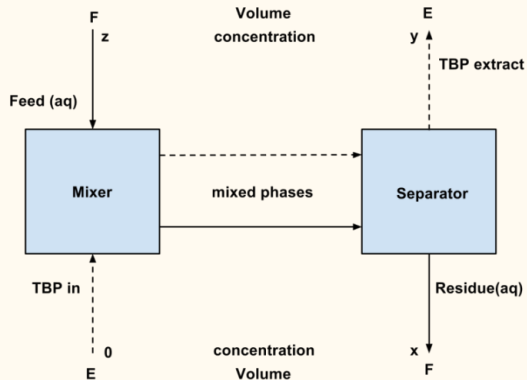
When an aqueous solution of an extractable component is brought into equilibrium with an immiscible solvent for the component and the two phases are then separated, the component will be found distributed between the two phases

## We then derive the material balance on the extractable component

$$F_z = F_x + E_y \quad (13)$$

With  $D$  for the system defined before

# PUREX single stage extraction



**Figure 64.** PUREX single stage extraction

## Derive the concentration that is extracted

$$D \equiv \frac{y}{x} \quad (14)$$

$$Fz = Fx + Ey \quad (15)$$

$$\therefore y = \frac{Dz}{1 + \frac{ED}{F}} \quad (16)$$

And the fraction extracted is –

$$\rho = \frac{Ey}{Fz} = \frac{1}{1 + \frac{F}{ED}} \quad (17)$$

For design then, you only need 3 parameters



## Derive the solvent feed needed for a given extraction fraction

$$\frac{E}{F} = \frac{\rho}{D(1 - \rho)} \quad (18)$$

What is the functional dependence of  $E(\rho)$ ?

Amount of used fuel isn't actually a variable

# Reduce the amount of solvent by using multiple contact stages

With many stages, high extraction fraction can be obtained

Solvent is expensive, so recycling stages are also added to strip out extract

Multiple components can be extracted from organic phase if  $D$  is different enough

So this would be needed to completely separate U, Pu

Physical conditions affecting  $D$  –

- Element extracted

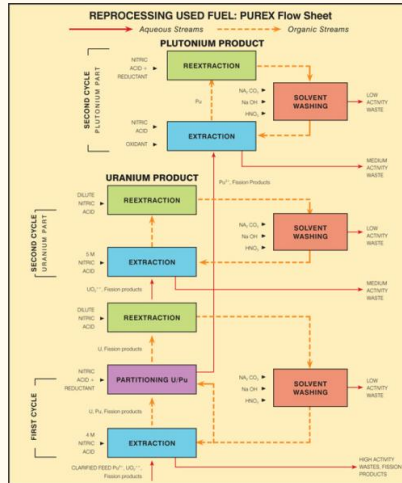
- Redox potential of aqueous phase

- Solvent

- Concentrations

- $[H^+]_{aq}$

# PUREX flowsheet



**Figure 65. PUREX flowsheet**

# Multistage extraction

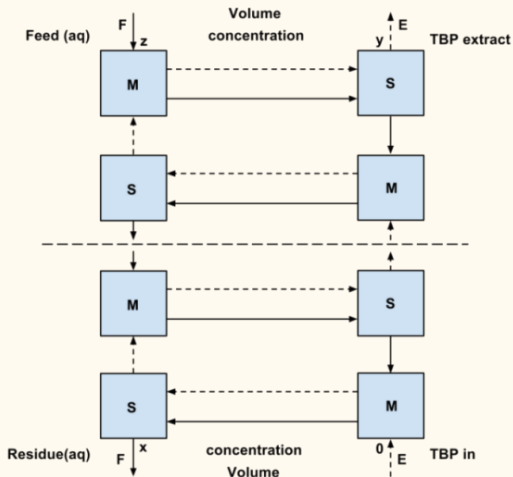
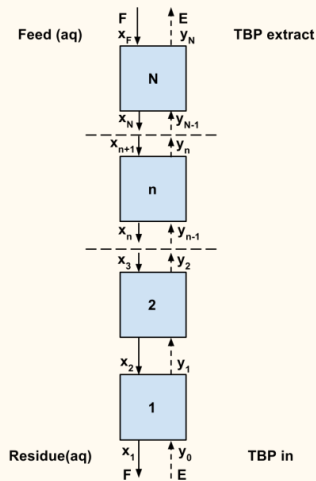


Figure 66. Multistage extraction

# Multistage extraction engineering diagram



**Figure 67.** Multistage extraction engineering diagram

## Now apply same mass conservation to multistage extraction

Assume equilibrium between phases

Material balance on extractable component below stage  $n$  –

$$Ey_0 + Fx_n = Ey_{n-1} + Fx_1 \quad (19)$$

$$y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1) \quad (20)$$

$\therefore$  for any stage –

$$y_n \equiv D_n x_n \quad (21)$$

## McCabe-Thiele diagram can be used for constructing graphical solution for stage concentrations

Basically gives the number of stages for an extraction concentration  $x_F$

By graphing  $y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1)$  on  $y$  versus  $x$

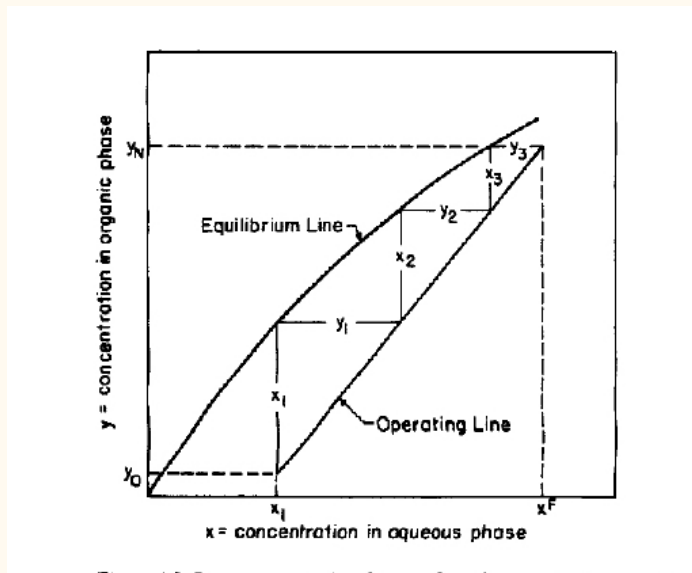
Construct operating line to satisfy the above and pass through  $x_1, y_0$  with slope  $\frac{F}{E}$  relative to equilibrium line  $y_n = D_n x_n$

Cannot really change equilibrium line because it is based on  $D$

Reduce the concentration of the extractable component from  $x_F$  to  $x_1$  with  $\frac{F}{E}$

Start at  $x_1, y_0$  and move up

# McCabe-Thiele diagram



**Figure 68.** McCabe-Thiele diagram



## Derive the extraction factor for multiple stages

$$\rho \equiv \frac{Ey_N}{Fx_F} \quad (22)$$

Or –

$$1 - \rho = \frac{x_1}{x_F} \quad (23)$$

What does this tell us?

And feed ratio can be defined in terms of  $f(\rho)$  as –

$$\frac{E}{F} = \frac{\rho x_F}{y_N - y_0} \quad (24)$$

**How many stages for complete extraction?**

**What does that look like on McCabe-Thiele?**

## Apply these concepts to model multistage extraction based on Kremser equation course notes

$$y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1) \quad (25)$$

$$y_n \equiv D_n x_n \quad (26)$$

$$\beta \equiv \frac{DE}{F} \quad (27)$$

Then –

$$y_n = \beta(y_{n-1} - y_0) + Dx_1 \quad (28)$$

$$y_N = \frac{\beta^N - 1}{\beta - 1}(Dx_1 - y_0) + y_0 \quad (29)$$

**This can also be applied if there are two extractable components in the feed**

$$D_A D_B \beta_A \beta_B \quad (30)$$

Define decontamination factor. What does this mean?

$$f_{AB} \equiv \frac{\rho_A}{\rho_B} \quad (31)$$

$$f_{AB} = \frac{\beta_A}{\beta_B} \left( \frac{\beta_A^N - 1}{\beta_B^N - 1} \right) \left( \frac{\beta_B^{N+1} - 1}{\beta_A^{N+1} - 1} \right), y_0 = 0 \quad (32)$$

## **Conversion & Enrichment**

# Uranium enrichment derived from World War II

Needed to make bombs

Then used the uranium for submarines

When moving to commercial power, infrastructure already in place

Gaseous diffusion first invented at Oak Ridge

Uranium hexafluoride used for enrichment

# Yellowcake needs to be purified first

Impurities are rare earths, chlorine cadmium

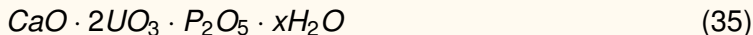
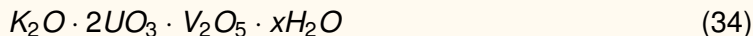
Uranium readily forms chemical compounds (lots of oxidation states)

Can be extracted by organic solvents

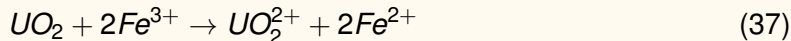
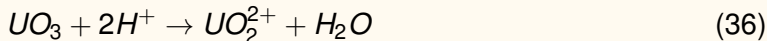
Uranium forms complexes that can be precipitated out



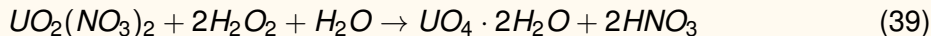
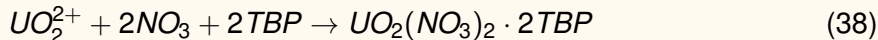
# There are different forms that are mined



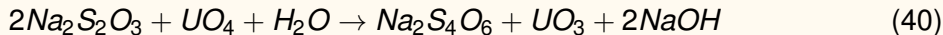
Goal is to get it as  $\text{UO}_3^{2+}$  and then acid or alkaline leaching



## Acid leaching is like PUREX using organic solvent

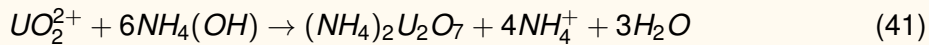


Precipitates uranium peroxide



Or use ammonium hydroxide

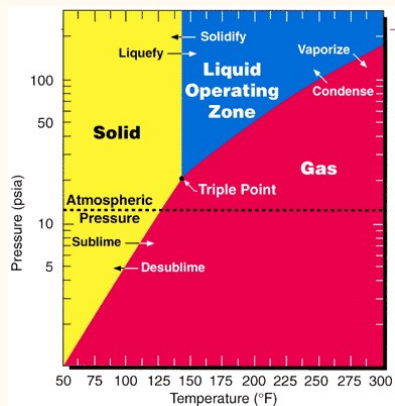
## Acid leaching is like PUREX using organic solvent



Then dry and calcine to get  $UO_3$

If alkaline leaching is used, recover  $U_3O_8$

# Uranium hexafluoride has favorable properties



Solid at room temperature

Easy to handle

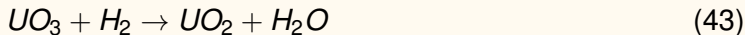
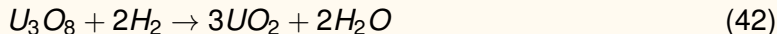
Relatively low pressure

Not too high temperature to sublimate

**Figure 69.** Uranium hexafluoride phase diagram

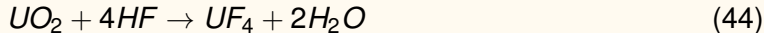
# Conversion applies hydrofluor process

Crushed  $U_3O_8$  and  $UO_3$  is reduced by hydrogen to get 'crude'  $UO_2$



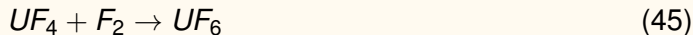
## Conversion applies hydrofluor process

Then –



$UF_4$  is a solid green salt with a high melting point ( $1700^\circ F$ )

Treated with fluorine gas and distilled –



# Enrichment based on different diffusion rates

Separation factor –

$$\alpha \equiv \frac{v_5}{v_8} = \sqrt{\frac{m_8}{m_5}} \quad (46)$$

Based on kinetic energy of the molecule  $kT = \frac{1}{2}mv^2$

This just says lighter molecules move faster

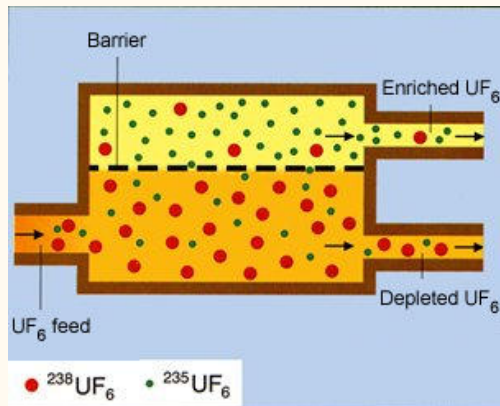
$\alpha = 1.004289$  for uranium enrichment

Higher value, easier enrichment

This isn't that high

Low enrichment takes a lot of energy

# Cascades are designed to enrich

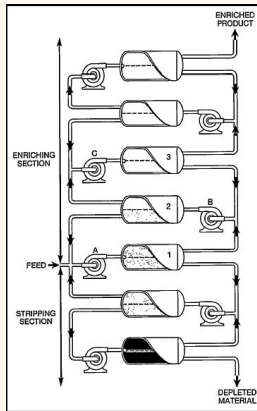


**Figure 70.** Centrifuge diffusion

- Basic operation is the stage
- Join stages to form cells
- Cells form unit
- Together all that is a cascade



# Enriching and stripping stages added for efficiency



Feed enters to A-1

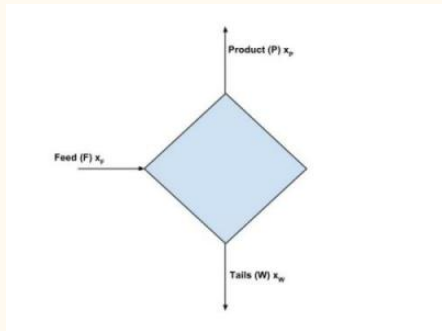
Enriched feed goes through B-2

Depleted through the bottom stage

Moving up is higher enrichment

**Figure 71.** Gas diffusion cascade

# Enrichment based on mass conservation



**Figure 72.** Mass conservation

$$F = P + W \quad (47)$$

$$x_F F = x_P P + x_W W \quad (48)$$

Separation factor ratio between product and tails –

$$\alpha \equiv \frac{x_P(1 - x_P)^{-1}}{x_W(1 - x_W)^{-1}} \quad (49)$$

## And mass flow

$x_F$  = weight fraction  $^{235}\text{U}$  feed into system

$x_P$  = weight fraction  $^{235}\text{U}$  product (enrichment target)

$x_W$  = weight fraction  $^{235}\text{U}$  waste stream (depleted U)

$F$  = mass flow rate feed

$P$  = mass flow rate product

$W$  = mass flow rate waste stream

# Derive feed factor

$x_F$  fixed at 0.00711

$x_P$  enrichment target

$x_W$  0.002 - 0.003 depends on company

Feed factor gives mass of U needed to feed into system for target enrichment –

$$\frac{F}{P} = \frac{x_P - x_W}{x_F - x_W} \quad (50)$$

## Derive feed factor

Similarly waste factor –

$$\frac{W}{P} = \frac{x_P - x_F}{x_F - x_W} = \frac{F}{P} - 1 \quad (51)$$

**How are tails managed?**

# Only one company enriches by gaseous diffusion

1992 – United States Enrichment Corporation established under Energy Policy Act

1998 – privatized with so you can buy stock

Operates plant in Paducah up to 5%

## Enrichment also done by centrifuge

1970 – URENCO established with United Kingdom, Dutch, Germany

1977 – Construction of US plant in Ohio, but Reagan scrapped it in favor of laser

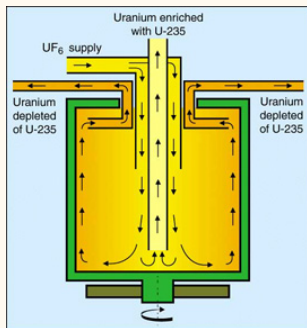
Rotating drum causes centrifugal force to compress gas molecules to the outer wall

Velocity of molecules separates differing those of differing weights

Light molecules collect near center of drum



# Still based on conservation of mass



**Figure 73.** Gas centrifuge

Lighter  $U^{235}$  molecules collect toward center

Rotational motion of the centrifuge accelerates gas molecules toward wall

Counteracting diffusive force thermal motion of molecules seeks to distribute the gas evenly

Two forces balance to create a dynamic equilibrium

Pressure distribution in the rotor that is function of the molecular mass

$$\alpha = \exp \frac{(m_8 - m_5) \omega^2 a^2}{2RT} \approx 1 + \frac{(m_8 - m_5) \omega^2 a^2}{2RT} \quad (52)$$

$$p(r) = p_0 \exp \frac{m \omega^2 r^2}{2RT} \quad (53)$$

## Design based on separative power

$$SWU = \frac{\pi \rho D L}{2} \left( \frac{\Delta m \omega^2}{2RT} \right)^2 \quad (54)$$

density of uranium hexafluoride; isotope self-diffusion; mass difference; length of rotor

Proportional to length and 4th power of speed

Inverse to temperature squared

What does this say about how to design the centrifuge?

It's actually small SWU per centrifuge, so up to 100,000 needed for 9 million SWU/y

That would be needed for industrial scale

# There are only 3 companies in the USA interested in centrifuges

USEC Inc. (USEC)

Louisiana Energy Services (LES)

AREVA Enrichment Services (AES)

USEC submitted application to operate a commercial facility

American Centrifuge Plant (ACP) August 2004

NRC issued a license April 2007

Construction of the facility is currently on hold (did not get DOE grant)

Chapter 11 in 2014 – now called [Centrus Energy Corp](#)

# Louisiana Energy Services activity

LES submitted license application NRC December 2003 for a 3 million SWU/year facility

One-third of the United States demand

Lea County, New Mexico

June 2006 – NRC issued LES a license to construct and operate

Began operations in June 2010 while continuing construction

Part of [URENCO](#)

July 2014 – All cascades operational

# AREVA Enrichment Services (AES) activity

December 2008 – AES submitted a license application for Eagle Rock Enrichment Facility to NRC

18 miles west of Idaho Falls!

September 2010 – NRC issued the [Safety Evaluation Report \(NUREG-1951\)](#)

February 2011 – [Final Environmental Impact Statement \(NUREG-1945\)](#)

October 2011 – NRC [issued AES license to construct and operate the Eagle Rock Enrichment Facility](#)

[Construction currently inactive status update](#)

## Lasers can be used for enrichment



Figure 74. Shark with laser beam

# Is laser enrichment any good?

Started in 1985

Absorption spectrum of uranium metal vapor

More than 300,000 lines at visible wavelengths

Sufficient displacement between lines

Selective excitation of  $^{235}\text{U}$  based on laser wavelength

Requires liquid uranium at  $2300^{\circ}\text{C}$

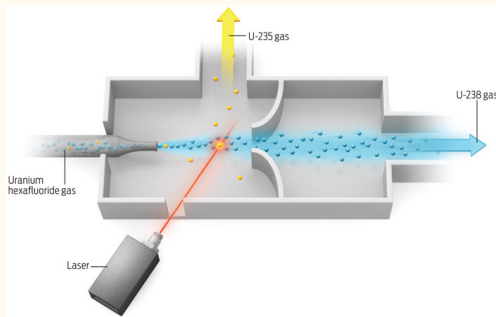
Laser at 5915 angstroms to excite  $^{235}\text{U}$

Ultraviolet light at 3100 angstroms ionizes  $^{235}\text{U}$

Then it is collected in a Faraday cup

Low throughput, high temperature, high energy

# Current developments



**Figure 75.** Laser enrichment

GE got an operating license

Doesn't seem like much progress, limited testing with DU in Paducah

Different process than what we were just talking about though

Looking to construct in Wilmington



# Han shot first!



**Figure 76.** Han Solo shoots Greedo

# Molecular separation uses uranium hexafluoride

Uses different vibrational and rotational states between isotopes

Almost the same procedure

$^{235}\text{U}$  molecule is dissociated and precipitates to a  $^{235}\text{UF}_5$  powder

Need to reduce the vapor pressure to 77K

# Underrated 80s gem



**Figure 77.** Real Genius

# SILEX seems to be state of the art

Uses carbon dioxide laser at 10.8  $\mu\text{m}$  and amplified to 16  $\mu\text{m}$

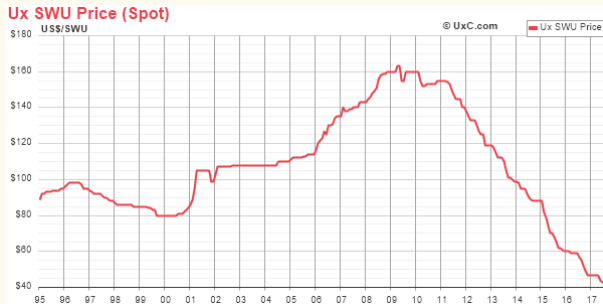
Infrared spectrum

Preferentially excites  $^{235}\text{UF}_5$

Again trapped electromagnetically

Still inefficient due to laser frequency

# Price used to be set by government



**Figure 78.** SWU price

From Atomic Energy Act 1954

Prices then set by USEC based on market conditions

Used to be high because of high energy requirements of gaseous diffusion plants

What now?

# World capacity is estimated at 60 million SWU

World enrichment capacity

US demand at 15 million SWU per year

Import a lot from Europe, Russia, China

What's the problem here?



# Acronyms I